

When the Turkish economy is in periods of expansion and recession, how is youth unemployment affected?

Muhammet Arif Kızıl

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Introduction

The Turkish economy has been facing economic difficulties and social problems for many years due to youth unemployment. Periods of economic expansion and recession in Türkiye also affect the labor market. When the economy slows down, companies usually reduce hiring or prefer experienced workers instead of younger employees with limited work experience.

Young unemployment is important because it distances young people from financial independence and, as a result, has a significant impact on the future of the country. Young people generally have less work experience than those who have been in the workforce for a long time. As a result, periods of economic downturn can affect young workers more harshly than other age groups.

The main research question of this project is the following:

Do economic expansions and recessions in Türkiye affect youth unemployment differently?

In response to this research question, the hypothesis tested in this project is the following.

Economic growth reduces youth unemployment, while recessions increase it.

Data

This project used macroeconomic data for Türkiye covering the years 1991–2024. The data used in the analysis was obtained from the World Bank database. Three key variables were used in the project: GDP growth, total unemployment, and youth unemployment.

The GDP growth data was used to analyze periods of expansion and recession in the Turkish economy. Total unemployment data was also used in the research to gain a better understanding of the labor market situation in Türkiye.

Before beginning the research, the data was cleaned and organized using Jupyter Notebook. The data was filtered to include only Türkiye, unnecessary observations were removed, and the data was combined into a single dataset for statistical analysis and visualization.

Methodology

This research examines the link between youth unemployment and periods of economic expansion and recession in Türkiye. The analyses were made using the Python programming language within a Jupyter Notebook.

First, the data was filtered to show only Türkiye, and the datasets were cleaned. Next, the data on GDP growth, total unemployment, and youth unemployment were combined. Summary statistics were used to analyze the data.

The research also used correlation analysis to examine the relationship between GDP growth and youth unemployment. Additionally, a linear regression model was used to test whether economic growth affects youth unemployment.

$$Youth\ Unemployment = \alpha + \beta(GDP\ Growth) + \varepsilon$$

Finally, graphs were prepared to show how economic variables have changed over the years. In order to compare how youth unemployment has changed under different economic circumstances, the data was also divided into periods of expansion and recession.

Results

In this section, the project's results are examined using statistical analyses and graphs. The main objective of the research is to examine the impact of periods of economic expansion and contraction on youth unemployment in Türkiye.

Summary Statistics

First, the summary statistics were used to investigate the overall nature of the variables included in the analysis. The results reveal that GDP growth rates in Türkiye have shown significant fluctuations over the years. During the period under review, Türkiye went through periods of strong growth as well as periods of economic crisis. The results also show that youth unemployment rates have stayed high for many years. Compared to total unemployment, the rates of youth unemployment are generally higher. This shows that young people may be more affected by economic fluctuations.

Table 1: Summary Statistics

	GDP Growth	Total Unemployment	Youth Unemployment
Count	34.00	34.00	34.00
Mean	4.63	9.82	18.43
Std. Dev.	4.37	1.90	3.13
Minimum	-5.46	6.50	12.94
25th Percentile	2.26	8.53	16.10
Median	5.27	10.06	18.37
75th Percentile	7.58	10.88	20.19
Maximum	11.81	14.03	25.09

Correlation Analysis

A correlation analysis was performed to investigate the relationship between GDP growth and youth unemployment. According to the results, the correlation value between the two variables was found to be quite low. This shows that there is no strong linear relationship between GDP growth and youth unemployment. This result shows that youth unemployment in Türkiye is not only affected by economic growth. Other factors, such as labor market conditions, educational level, lack of work experience, and economic uncertainties, can also influence youth unemployment.

Table 2: Correlation Matrix

	GDP Growth	Total Unemployment	Youth Unemployment
GDP Growth	1.000	-0.023	0.008
Total Unemployment	-0.023	1.000	0.988
Youth Unemployment	0.008	0.988	1.000

The correlation between GDP growth and youth unemployment is 0.008, which is almost zero. This means that there is no strong linear relationship between economic growth and youth unemployment in this dataset.

Regression Analysis

Table 3: Regression Results

Variable	Coefficient	p-value	R-squared
GDP Growth	0.0059	0.963	0.000

A linear regression analysis was conducted to see if there is any impact of GDP growth on youth unemployment in Türkiye. According to the analysis, the GDP growth coefficient is extremely low at 0.0059. Also, the p-value of GDP growth is 0.963, which is much larger than 0.05, the generally accepted p-value. Hence, GDP growth is not statistically significant in relation to youth unemployment.

R-squared is close to zero, which means that GDP growth does not have much explanatory power regarding youth unemployment.

In general, the regression analysis shows that youth unemployment is determined not only by the economic growth but also other factors. These factors could include labor market conditions, level of education and work experience. In other words, the solution of the problem of youth unemployment cannot be reduced only to the growth of economy.

Graphical Analysis

Graphs were used in order to have a better insight into the relationship between the growth of GDP and youth unemployment in Türkiye. Below are presented the graphs that will serve as a representation of the data analyzed statistically.

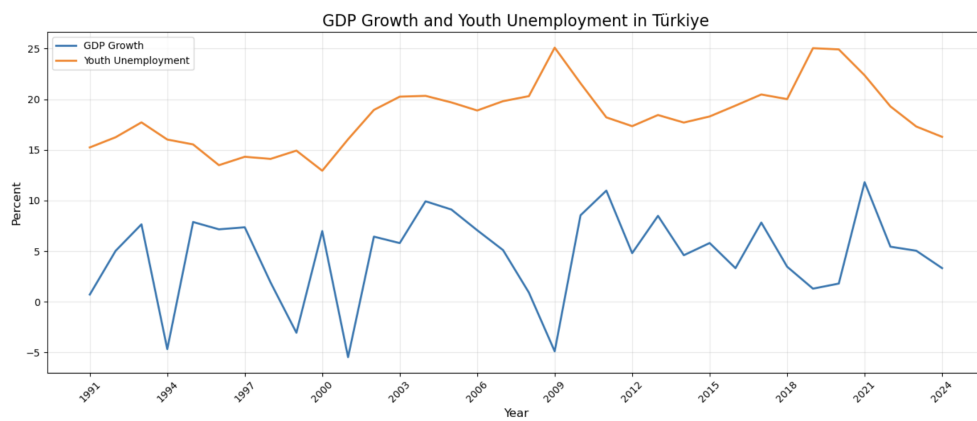


Figure 1: GDP Growth and Youth Unemployment in Türkiye

The dynamics of GDP growth and youth unemployment are presented in Figure 1. It can be seen from the picture that there was much fluctuation regarding GDP growth in the period under study. On the other hand, youth unemployment was not always changing in a reverse pattern to GDP growth. It proves the results obtained in the regression analysis.

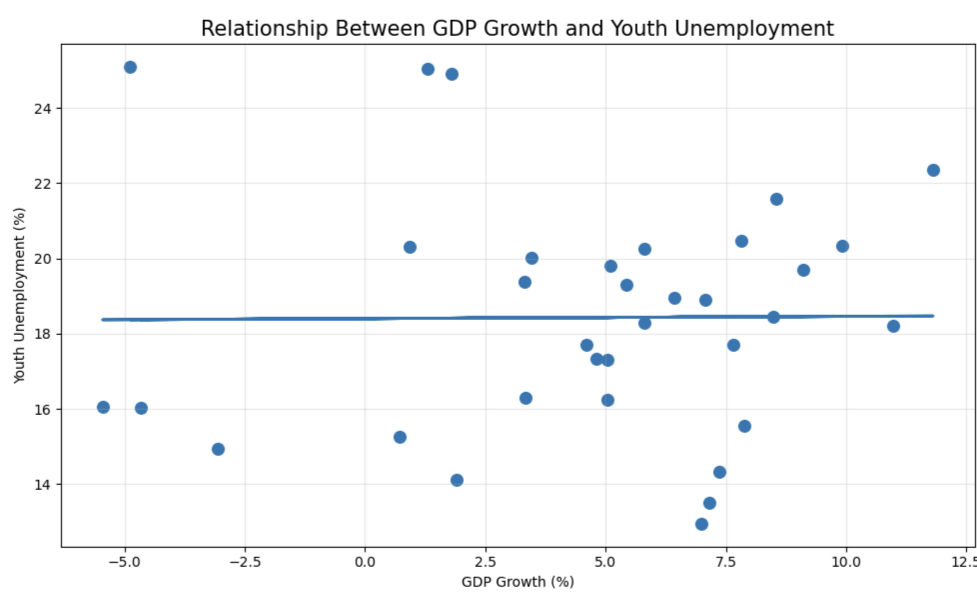


Figure 2: Relationship Between GDP Growth and Youth Unemployment

Graph 2 represents the relationship between economic growth and youth unemployment. The data points are scattered and show no recognizable pattern. In addition, the regression line is nearly horizontal. Therefore, it can be seen that the relationship between

economic growth and youth unemployment is relatively weak, which is confirmed by the correlation and regression analysis.

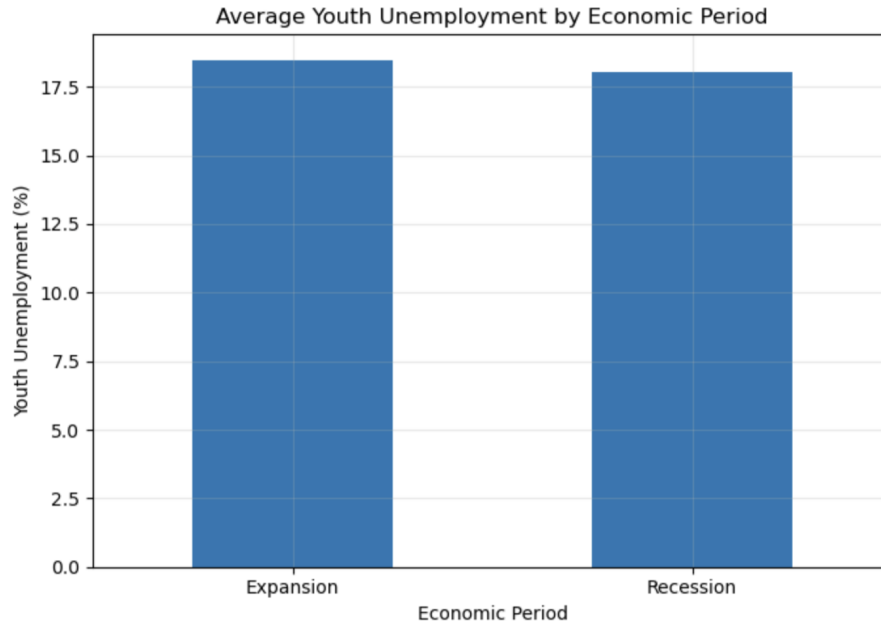


Figure 3: Average Youth Unemployment During Expansion and Recession Periods

Figure 3 illustrates youth unemployment compared between times of economic growth and recession periods. Youth unemployment is seen to be relatively higher during recession periods. However, there is a small margin of variation, which implies that youth unemployment is a relevant concern even during economic growth periods.

Conclusion

This study analyzed the relationship between economic expansion, recession, and youth unemployment in Türkiye from 1991 to 2024. According to the findings, the level of youth unemployment was quite high during all the years under consideration, including economic growth years. Based on the results of the correlation and regression analyses, it can be stated that GDP growth does not have a strong influence on the reduction of youth unemployment in Türkiye. The correlation between GDP growth and youth unemployment turned out to be quite low. Moreover, the regression analysis did not produce statistically significant results. Besides, comparing the periods of expansion and recession, it becomes evident that the period of economic recession is associated with higher youth unemployment. However, the gap between the two periods was not too big. Therefore, it can be concluded that GDP growth alone cannot lower the level of youth unemployment in the country.